



RV COLLEGE OF ENGINEERING®
Department of Computer Science and Engineering
CIE-1: Question Paper

Course : (Code)	PARALLEL ARCHITECTURE & DISTRIBUTED PROGRAMMING (21CS72)	Semester : 7
Date : Dec 2024	Duration : 120 minutes	Staff : MMN/ MM /AN
Name :	USN :	Section : A/B/C

Answer all questions

Sl.no	Questions	Marks	L1-L6	CO
1	Imagine you are using a web browser to navigate a website. The browser accesses data such as HTML files, CSS stylesheets, JavaScript scripts, and images to render a webpage. For the following scenarios, identify type of locality: i. If you visit a news website, the browser might reuse the cached logo image for every page on the site. ii. If you're reading an online article, the browser preloads text, images, and videos located in the next section of the article.	2	1	1
2	A program has 80% of its workload parallelizable ($P=0.8$) and is executed on 4 processors ($N=4$). What is the speedup?	2	2	2
3	For the following code snippets identify the type of hazards: i. ADD R1, R2, R3 MUL R4, R5, R6 ii. LW R1, 0(R2) SW R3, 4(R4)	2	1	1
4	Simulating weather for a country using a 1000×1000 grid, where each grid cell represents a 1 km^2 . Identity the gran size.	2	2	1
5	Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate-Pipeline cycle time.	2	2	1

Part B

Sl.no	Questions	Marks	L1-L6	CO
1. a.	List and explain the Quantitative Principles of computer design.	6	L2	C
b.	Assume a server subsystem with the following components and MTTF values: 8 CPUs, each rated at 2,000,000-hour MTTF 2 power supplies, each rated at 1,500,000-hour MTTF 4 memory modules, each rated at 500,000-hour MTTF 2 cooling fans, each rated at 1,200,000-hour MTTF 1 network card, 3,000,000-hour MTTF Using the simplifying assumptions that the lifetimes are exponentially distributed and that failures are independent, compute the MTTF of the entire server subsystem.	4	L3	C

2	Show how the loop would look on MIPS. both scheduled and unscheduled including any stalls or idle clock cycles. Schedule for delays from floating point operations but remember to ignore delayed branches for(i=1000;i>0;i=i-1) x[i]= x[i]+s ;	10	L2	CO2
3 a.	Differentiate between UMA memory and NUMA memory architecture.	6	L3	CO2
b.	Demonstrate with suitable example the cache coherence problem for multiprocessor architecture.	4	L4	CO1
4	Draw and explain the 5-stage pipeline in a processor. Define control hazards and describe strategies to avoid them.	10	L2	CO2
5 a.	Illustrate Tomasulos algorithms with neat diagram	10	L3	CO2

CO1: Apply parallel programming techniques to investigate parallel algorithms for solving computational problems.

CO2: Analyze the performance of parallel algorithms on different parallel architectures.

CO3: Design and implement parallel algorithms using appropriate parallel programming models such as Open MP, MPI, CUDA and OpenACC

CO4: Demonstrate Parallel computing concepts for suitable compute intensive real time applications

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4
Marks	4	33	18	5	-	-	19	41		